17/9/2025

Task-9.1

Implement Exceptions and exceptional handling in Python.

Aim: To implement Exception and Exceptional handling in python.

Algorithm:

1) Start the program
2) Initializes a list of grades
3) prompts the user to enter the index of the grade
4) Attempts to dissplay the grade at the index
5) If the index is out of range, catches the index error.

Program:

```
#initialize the list of grades
grades=[85,90,78,92,88]
#Display the grades list
print("Gades list:", grades)
#' promt the user to enter the index of the grade
they want to view
Index= int(input("Enter the index to view:"))
# Attempt to display the grade
printf("The grade at index
Except index Error:
#Handle the case where the index
Print("Invalid error.
# Handle the case where the input
print("Invalid error]
```

Output:

Grades list : [85,90,78,92,88]

Enter the index of the grade you want to view : 10

Invalid error : please enter a valid index.

Enter the Numerator = 10

Enter the denominator = 0

Error!

Error : Division by zero is not allowed.

Task-9.2

Aim: To develop a python calculator program that performs basic arithmetic operations.

Algorithm:

1) Start the program
2) prompts the used to enter two numbers
3) Attempts to divide the numerator by the denominator
4) If the denominator is zero, catches the zero Division Error and displays an error message.

Program:

```
# Function to perform division
def divide_numbers():
    try:
        # prompt the user to enter
        numerator = float
        # prompt the user to enter
        denominator = float
        # Attempt to perfor division
        result = numerator / denominator
        print("result is {result}")
    Except zero division error:
    # Handle division by zero error.
        print("Error")
    Except value error.
        # Handle invalid input.
        print("Error.")
# call the function to execute.
    divide_numbers()
```

Output:

```
Enter a number: 15
Exception occurred = Invalid Age
```

Result:

Task-9.3

Aim: To building a python application to determine if a person is eligible to vote based on their age.

Algorithm:

1) Define the custom Exception
2) prompt the user for input
3) check if the age is below 18.
4) Raise an exception.
5) Handle the exception with a custom.

Program:

```
# define python user-defined exceptions
class invalid Age Exceptions
"Raised when the input value
# you need to guess this number
   number = 18
Try:
 Input_num = int(input(""))
 If intput_num < number:
  raise invalid age exception
 else:
  print("Elgible to vote")
Except Invalid Age exception
 print("Exception occured
```

Result: Thus the program for implement exceptions and Exceptional handling is Executed and verified Successfully

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RESULT AND ANALYSIS (5)	
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